JOBHELPER

# JobHelper technical documentation

How the app is built, run and tested

For JobHelper on macOS
10 figures · 23 tables

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**Applies to:** the JobHelper source code, on macOS with Python 3.10 or newer.

This document is for people who work with the code: it explains how to install the app, how to run it from the command line, how the code is laid out, and how to test and document a change. To use the app, read the user guide, GUIDE.md.

# 1. Overview

JobHelper is a desktop app for macOS that finds job postings, scores them against a CV, writes an adapted CV and a cover letter for the good ones, and tracks applications and replies. It is written in Python. The window is built with Qt Quick (PySide6). All model calls go through the Claude Code program, so that they use the person's Claude subscription and never the paid API.

The work is done by five agents (Figure 1.1). A search runs the first four in order; an email scan runs the fifth.

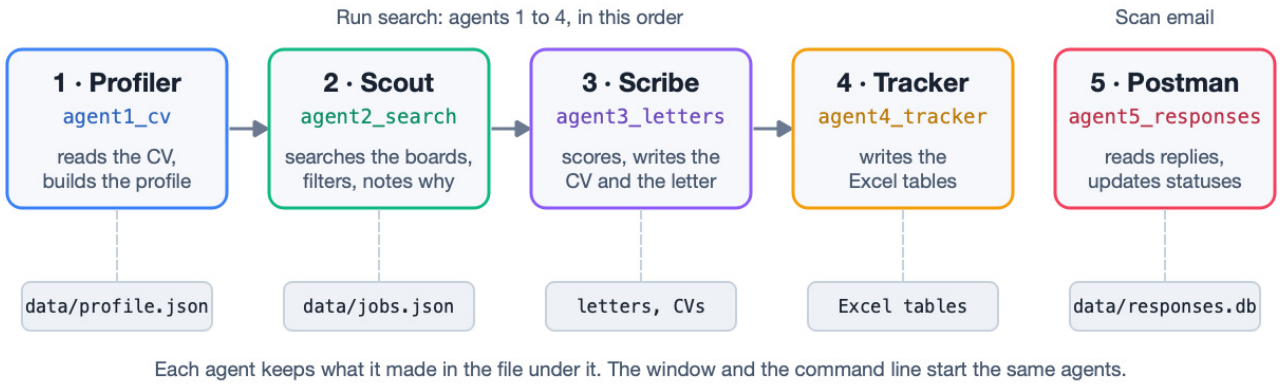



Figure 1.1 – The five agents and what each keeps

Six rules shape the whole code base, and every change must keep them (Table 1.1).

Table 1.1 – The rules every change must keep

| Rule | What it means in the code |
|---|---|
| The subscription, never the paid API | llm.ask refuses to run while ANTHROPIC_API_KEY is set. No code path calls the API directly |
| The mailbox is read-only | Folders are opened read-only and messages are fetched with PEEK. Nothing sends, deletes, moves or marks mail |
| Nothing is claimed that the CV does not show | Every written text is checked in plain code against the CV after the model wrote it |
| The person's own letters are never changed | Only letters the app wrote may be replaced or moved |
| Nothing is deleted | Data is backed up before a bulk change; documents are moved to an encrypted backup, old copies to the Trash |
| No personal data in the code | Names and facts of a real person never appear in code, tests or documents |

### Related documents

Table 1.2 lists the other documents of the project.

Table 1.2 – The documents of the project

| Document | Use it for |
|---|---|
| GUIDE.md | Using the app, for someone who never opens a terminal |
| COMMANDS.md | The full reference of every command and option |
| HANDBOOK.md | The rules the app is built by, decisions already made, what is left to do |
| CHANGELOG.md | What changed, day by day. The app shows it under **Help** |

## 2. Know the tech stack

### Overview

JobHelper is built from a small number of well-known parts, each chosen for a reason that follows from what the app must do: run on one person's Mac, keep their

data private, and use their Claude subscription instead of a paid API. This chapter names each part and says why it was chosen, so that a change does not replace one without knowing what it was there for.

## The language and the window

Table 2.1 lists what the app itself is written in.

Table 2.1 – The language and the window

| Technology | Used for | Why it was chosen |
|---|---|---|
| Python 3.10 or newer | All the logic | The libraries for Word files, Excel files, mail and web pages are mature in Python, and the official Claude library is a Python library. One language for everything keeps the project small enough for one person |
| Qt Quick (QML) with PySide6 | The window | It draws a native window on the graphics card, so the robots and the lists animate smoothly, and it needs no web browser inside the app. An earlier window built from a web page was removed: two front ends had to be kept the same, and the web one was slower and looked foreign on macOS |
| QML for layout, Python for logic | The split between ui_qt/ and jobhelper/ | A QML file only shows things and calls the bridge; it never reads a file. The logic can therefore be tested without a window, and the window can be tested without real data |
| Threads with Qt signals | Searches, scans and rewrites | Long work must not freeze the window. A worker thread does it and answers through a signal, which Qt delivers on the window's own thread |

## Claude

Table 2.2 lists how the app reaches the model.

Table 2.2 – How the app reaches Claude

| Technology | Used for | Why it was chosen |
|---|---|---|
| claude-agent-sdk driving the Claude Code program | Every question to the model | It uses the person's Claude subscription. The paid API bills per use and would make every search cost money; that is why the app refuses to run while ANTHROPIC_API_KEY is set |
| One module, llm, for all calls | Asking, counting usage, noticing the limit | With a single door, the rules hold everywhere: no call while the limit is used up, every call counted, no key in an error message |
| Plain code before and after the model | Filters before scoring; checks after writing | A check in code is free, instant and always gives the same answer. The model is asked only what code cannot decide, and what it wrote is checked against the CV in code |

### Documents and data

Table 2.3 lists what reads and writes the person's files.

Table 2.3 – Documents and data

| Technology | Used for | Why it was chosen |
|---|---|---|
| python-docx | Reading a CV; writing the adapted CV and the letter | Employers expect Word files. Editing the person's own .docx keeps its fonts and layout, where generating a new file would lose them |
| pypdf | Reading a CV that is a PDF; joining the letter and the CV into one PDF | It is pure Python, so nothing else has to be installed |
| LibreOffice (optional) | PDF copies of Word files | It converts a Word file faithfully and is free. It is optional: without it the app still works and offers Word files only |
| openpyxl | The Excel tables | People already track applications in Excel; the tables can be opened, sorted and shared without the app |
| JSON files | The job list, the profile, the Options | The data of one person is small. A file can be read whole, backed up by copying, and repaired by hand if it must be |
| SQLite | The replies read from the mailbox | Replies are many, are looked up by message, and must never be read twice. A small database does that; it is part of Python and needs no server |

### Reading the web and the mailbox

Table 2.4 lists what reaches outside the Mac.

Table 2.4 – Reading the web and the mailbox

| Technology | Used for | Why it was chosen |
|---|---|---|
| requests with beautifulsoup4 and lxml | Reading the job boards and company career sites | The boards have no common interface, so their public pages are read as a browser would read them, at a modest pace. No search needs a key or an account |
| imaplib, from Python itself | Reading replies | IMAP works with almost every mail provider and can open a mailbox read-only, which is the promise the app makes: it never sends, deletes or marks mail |
| An app password instead of a sign-in window | Access to the mailbox | It can be cancelled at the provider at any time without changing the real password, and it needs no registration of the app with each provider |

### Privacy and the Mac

Table 2.5 lists what protects the data and ties the app to macOS.

Table 2.5 – Privacy and the Mac

| Technology | Used for | Why it was chosen |
|---|---|---|
| AES-256-GCM, from cryptography | Encrypting the data files and the backups | Applications and replies are private. This method also notices when a file was changed, so a damaged or tampered file is refused instead of read |
| The macOS Keychain, through the security program | Keeping the data key and the mailbox passwords | Secrets are then protected by the Mac's login and are never written to a file of the project |
| Signed rule files | The rules for CVs and letters, the letter template, the master CV | These files decide what is written in the person's name. A change made outside the app must be approved before the next run |
| A LaunchAgent | The weekly run | It is how macOS starts a task on a schedule, also when the window is closed |
| Apple events, through the osascript program | The assistant reads and fills in the form that is open in the person's browser (Chrome, Brave, Edge, Opera or Safari; Opera GX has that door shut and Firefox never had it) | The form is filled in where the person is already signed in, with nothing to install. Chrome and macOS each ask for consent first, and the app can only do what a script in the page can do: it cannot attach files or read saved passwords |
| Hash-pinned packages in requirements.lock | Installing | Each package is checked against a fingerprint, so a changed package cannot be installed unnoticed |

### Quality

Table 2.6 lists what keeps the code correct.

Table 2.6 – Tools for quality

| Technology | Used for | Why it was chosen |
|---|---|---|
| pytest | More than 1,300 tests | Every change comes with tests. The suite needs no network and never touches real data, so it can be run at any time |
| The window loaded without a screen | Tests of the window | The real QML is loaded and driven, so a broken binding or a missing name fails a test instead of reaching the person |
| pyflakes | Unused and undefined names | It finds a whole kind of mistake in a second, before a test has to |
| Git, kept on the Mac only | The history of changes | Every finished step is a commit that can be gone back to. There is no remote copy, because the project holds a real CV |

### What is deliberately not used

Table 2.7 lists parts that were considered and left out.

Table 2.7 – What is deliberately not used

| Not used | Why not |
|---|---|
| The paid Claude API | It bills per use. The subscription the person already has does the same work |
| A web browser inside the app | It was tried and removed: slower, larger, and a second front end to keep the same |
| A browser extension for filling in forms | It would have to be installed by hand and kept in step with the app. Apple events reach the same page without one |
| A database server | One person's data fits in files; a server would be one more thing to install and to keep running |
| A cloud service for the data | Applications, replies and the CV stay on the person's Mac |
| Sending mail | The app reads replies and drafts texts. The person sends them, so nothing leaves in their name without them |

### Next steps

- Install the app

# 3. Install the app

### Prerequisites

Before you install, make sure you have:

- macOS.
- Python 3.10 or newer.
- The Claude Code program, signed in with a Claude subscription: run claude auth login once.
- Optional: LibreOffice, for PDF copies of the documents.

> **Important**
> Do not set ANTHROPIC_API_KEY. Leaving it unset is what makes the app use the subscription, and the app refuses to run a search while it is set.

### Install the packages

1. Open a terminal in the project folder.
2. Create a virtual environment and turn it on:

```
python3 -m venv .venv
```

```
source .venv/bin/activate
```

3. Install the exact, hash-checked packages:

```
pip install --require-hashes -r requirements.lock
```

4. Optional: copy the example settings file. The mailbox password never goes into it.

```
cp .env.example .env
```

Table 3.1 lists the main packages and what each is for.

Table 3.1 – The main packages

| Package | Used for |
|---|---|
| claude-agent-sdk | Driving the Claude Code program, which uses the subscription |
| PySide6 | The window (Qt Quick and QML) and the PDF of the documentation |
| python-docx, pypdf | Reading a CV; writing CVs and letters as Word files |
| openpyxl | The Excel tables |
| requests, beautifulsoup4, lxml | Reading the job boards |
| cryptography | AES-256-GCM encryption of the data files and the backups |
| pytest | The tests. pyflakes, installed separately with pip install pyflakes, checks for unused or undefined names |

# 4. Run the app

## Start the window

1. Turn the virtual environment on.
2. Start the app:

```
python app_qt.py
```

3. Follow the welcome card: sign in to Claude, choose a CV, and connect a mailbox if you want replies to be read. The user guide describes each step.

## Run a search without the window

A search can also be run from the command line. It does the same work as **Run search** and writes to the same files.

```
python run.py
```

Table 4.1 lists the options used most often.

Table 4.1 – The most used options of run.py

| Option | What it does |
|---|---|
| --where CITY | Searches in another city than the one saved in Options |
| --source NAME | Chooses the job boards: czech, linkedin, remote, adzuna, jooble or all |
| --include-remote | Adds the remote boards |
| --threshold N | Writes documents only from this score |
| --per-query N | Asks each board for this many postings per role |
| --all-languages | Keeps postings written in other languages than English |
| --max-age-days N | Drops postings older than this |
| --reuse-profile | Does not read the CV again |
| --skip-search | Scores the postings already in the list, without searching |

> **Note**
> Do not run a search from the command line while the window is running one. Both write the same job list.

## 5. Use the command line

Everything the window does with applications and replies can also be done with status.py. Table 5.1 lists its commands.

Table 5.1 – The commands of status.py

| Command | What it does |
|---|---|
| python status.py list | Lists every tracked job with its id and status |
| python status.py apply ID | Marks a new job as applied |
| python status.py update ID STATUS | Sets the status of a job |
| python status.py sync | Reads replies in the mailbox and updates statuses. --days N chooses how far back |
| python status.py responses | Shows the numbers of replies and rejection reasons |
| python status.py followups | Lists applications with no reply for a number of days |
| python status.py rebuild | Writes the Excel tables again |
| python status.py retailor ID | Writes a letter and an adapted CV for one job |
| python status.py pdf | Makes PDF copies of the documents (needs LibreOffice) |
| python status.py letters | Takes in letters the person wrote themselves |
| python status.py prune | Removes new jobs you have already applied to |
| python status.py repair-emails | Reads email text again, without Claude, and fixes replies that were misread because the email was HTML only |
| python status.py repair-statuses | Fixes jobs whose status the app does not know |
| python status.py usage | Shows the Claude usage of the app, in total and per kind of work |
| python status.py security ACTION | Encryption, Keychain and backup tools (Table 5.2) |

### Protect and restore the data

The data files are encrypted with a key kept in the macOS Keychain. Table 5.2 lists the actions of python status.py security.

Table 5.2 – The actions of status.py security

| Action | What it does |
|---|---|
| status | Says whether encryption is on and whether a watched file changed |
| recovery-key | Prints the recovery key. The window shows the same key in the **Account** card |
| restore-key | Puts a saved recovery key back into the Keychain |
| approve | Accepts a change to a rule file, the letter template or the master CV as trusted |
| backup, backup-all | Makes an encrypted backup of the data, or of everything |
| restore, test-restore | Restores a backup, or proves that a backup can be restored |

> **Warning**
> Anyone who has the recovery key can read the data. Never print it into a log, a chat or a file that is kept.

## 6. Understand the structure of the code

The code is in three places (Table 6.1).

Table 6.1 – The folders of the project

| Folder | What is in it |
|---|---|
| jobhelper/ | All the logic: the agents, the rules, reading and writing files |
| ui_qt/ | The window: one QML file per window or part of the main screen |
| tests/ | The tests, and the script that takes the pictures of the user guide |
| scripts/ | Tools for the documentation: the PDF builder, the numbering check, the diagram |
| docs/ | The user guide, this document, their PDFs and the pictures |
| data/ | The person's data. Never part of the code; see section 7 |

### The agents

Table 6.2 names the module of each agent and what it may ask Claude.

Table 6.2 – The five agents

| Agent | Module | What it does | Asks Claude |
|---|---|---|---|
| Profiler | agent1_cv | Reads the CV and builds the profile and the search roles | Once per CV |
| Scout | agent2_search | Searches the boards, filters, keeps every dropped posting with its reason | Never |
| Scribe | agent3_letters | Scores postings five per call; writes the adapted CV and the letter | Yes |
| Tracker | agent4_tracker | Writes the Excel tables | Never |
| Postman | agent5_responses | Reads replies and files each to its application | For new replies only |

## What a search does with a posting

Figure 6.1 follows one posting through a search. Everything before the scoring is plain code: a posting that does not fit the rules, or is plainly another profession, never costs a call to Claude, and it is kept with the reason so that the person can bring it back.

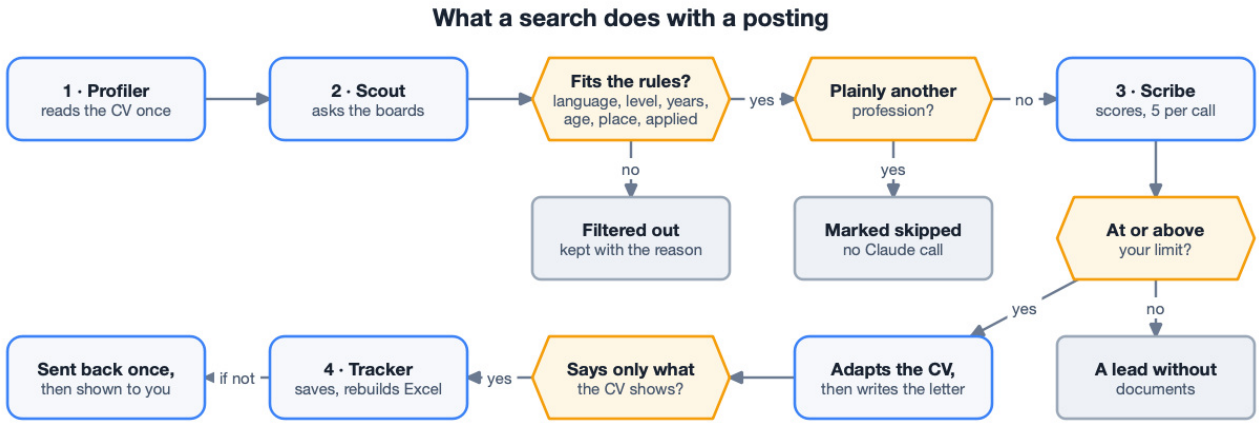



Figure 6.1 – What a search does with a posting

## How a text is kept inside what the CV shows

An adapted CV, a letter, a form answer and an interview answer are all written by Claude and then checked in code (Figure 6.2). A prompt that says "do not invent" is not a guarantee, so the draft is read for technologies the CV does not show and for terms of the posting stated as experience. What is found is sent back once; what is still there is kept and shown to the person, never cut out silently.

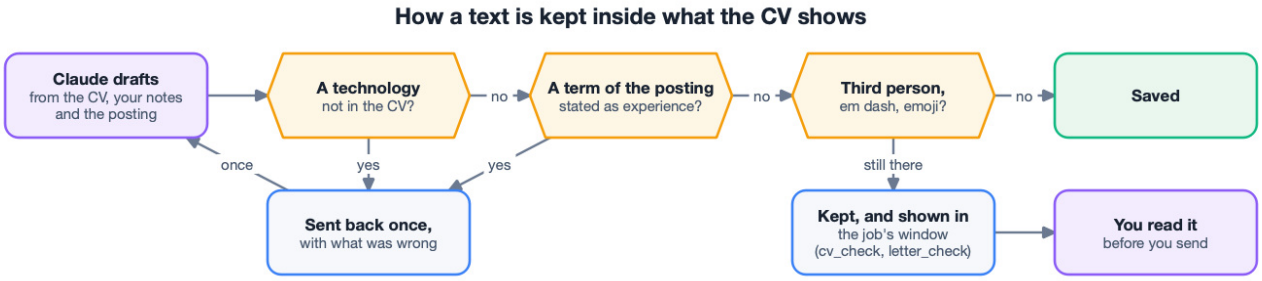



Figure 6.2 – How a text is kept inside what the CV shows

## What an email scan does with a message

Figure 6.3 shows the way of one message. The mailbox is opened read-only, and rules decide without a call which messages are worth a question to Claude.

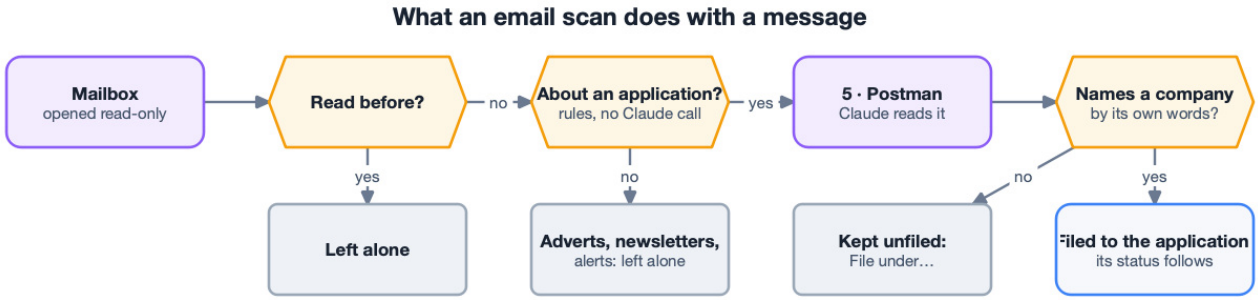



Figure 6.3 – What an email scan does with a message

## How a run behaves at Claude's limit

The limit of the subscription is a state of a run, not an error (Figure 6.4). Every call goes through llm.ask, which refuses to call when a limit is used up; the run saves its work, shows the reset time, and goes on from the same job when the person presses **Resume**.

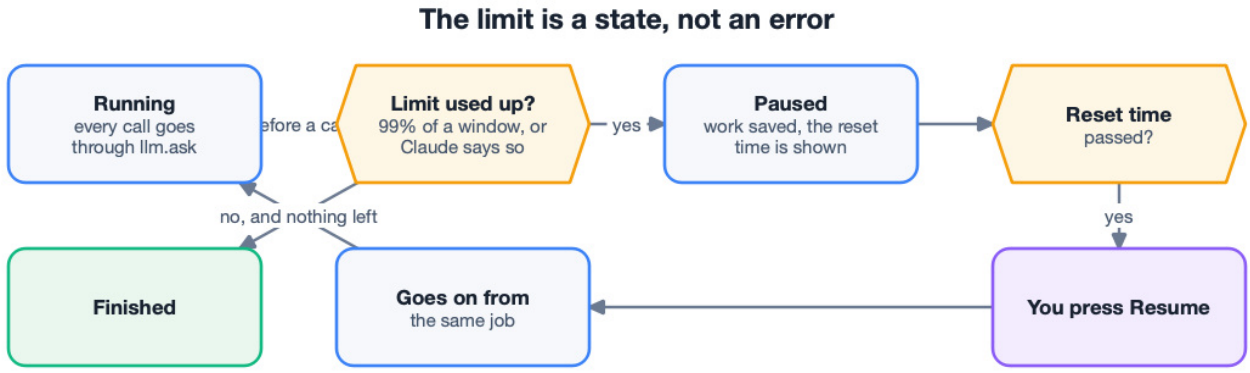



Figure 6.4 – The limit is a state, not an error

## The modules around the agents

Table 6.3 groups the other modules by what they are for.

Table 6.3 – The modules around the agents

| Group | Modules | What they do |
|---|---|---|
| Reading postings | scrapers_cz, scrapers_intl, scrapers_remote, scrapers_ats, scrapers_page | One module per kind of source. A site that turns programs away is left alone |
| Rules without a model call | filters, prefilter, language, places, keywords, salary, merge, followup, planner | Level, years, language, location, profession, pay, duplicates, reminders |

Continuation of Table 6.3

| Group | Modules | What they do |
|---|---|---|
| Writing | cv_tailor, cover_letter, letters, form_answers, interview_prep, prep_doc, pdf | The adapted CV, the letter, form answers, the interview sheet |
| Mail | email_client, mailhosts | Reading mailboxes read-only; which server a mailbox is read from |
| The assistant | assistant, assistant_brain, assistant_notes, details, chrome, form_fill, form_choices, answer_memory | The fixed requests and the facts about the app; what Claude makes of a free request, checked before anything happens; the person's own details; the link to Google Chrome; reading a form, deciding what each field is, and filling it in; choosing from lists and menus; the person's own wording, remembered |
| Claude | llm, usage | The one place that asks Claude; counting usage and knowing the limits |
| Storage and safety | store, db, vault, integrity, housekeeping, health, trouble | The job list, the reply database, encryption, signed rule files, backups, the start-up check, the log of errors |
| The window's back end | ui_api, api_documents, api_interview, api_letters, api_assistant, api_talk, qt_bridge, bridge_runs, bridge_jobs | What the window calls. Api holds the logic; Bridge exposes it to QML and runs long work in threads |

### How the assistant fills in a form

The assistant works on the tab that is in front in the person's browser (Figure 6.5), in steps that are each small enough to check, all in form_fill:

1. A script lists the fields of the page: what each is called, its kind, and whether it already holds something. It numbers them with an attribute and sends back no value the person typed.

2. plan() decides in plain code what each field is. The person's details come from details; open questions go to form_answers in one call; everything else is left, with the reason.
3. A second script writes the values into fields that are still empty and outlines them.
4. Lists, radio buttons and menus the page draws itself are choices: form_choices decides them, and a third script opens a menu, reads its entries, and presses the chosen one.
5. A file field that asks for the CV, the letter or other attachments (file_kind) is given the document of the job. The file travels to the page as text, in pieces of 150,000 characters (PUT), because a request to the browser has a size limit; ATTACH makes the pieces into a file and hands it to the field the way a file chosen by hand arrives.

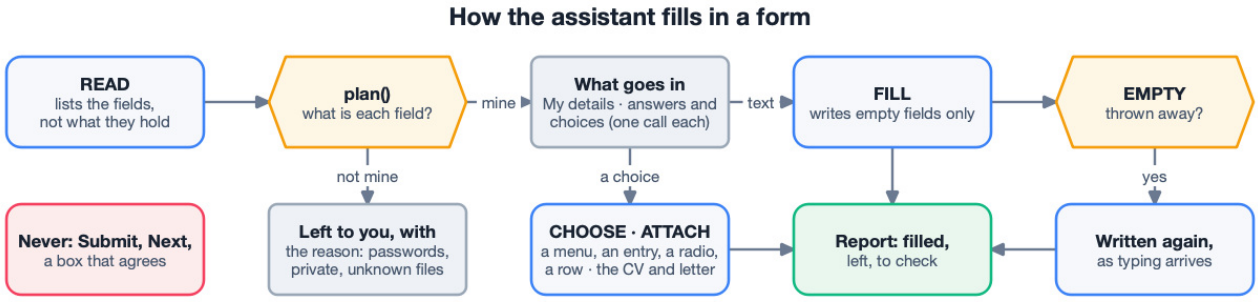



Figure 6.5 – How the assistant fills in a form

The job a form belongs to is found by match_job. A form is rarely at the address of its posting, so the function tries, in this order: the address of the posting (of the page, or of the page that led to it); the company's name in the address or the title; the company and the job title both in the words of the page; and the posting last opened from the app, when the page names its company or its title. Anything less plain gives no job, and without a job no file leaves the app.

Table 6.4 gives the rules these steps keep. tests/test_assistant.py checks each of them.

The whole request is also tried in a test area: tests/fixtures/fill_in_browser.py serves the forms in tests/fixtures/forms/ from this machine, opens one in a browser engine, and runs the same code the window runs, with Claude replaced by prepared answers and the data in a temporary folder. There is a plain form, one built the way component libraries build them (each box keeps its own value and undoes what did not arrive like typing; its lists are menus that hold nothing until they are opened), a long one with lists, radio buttons and a number, and one that sits in a frame from another address. To check a kind of form the assistant gets wrong, add a page there and a test that names what must be filled and what must be left.

Table 6.4 – The rules of filling in a form

| Rule | How it is kept |
|---|---|
| The form is never sent | No script presses a button that sends. The one script that presses anything (CHOOSE) accepts five kinds of element: what opens a menu, an entry of a menu, a radio button, a button whose words say that it adds a row for a job or a degree, and a tick box whose words say that a job goes on. It reads those words again before it acts, and refuses anything that sends, continues, uploads or agrees. A test reads the scripts for those calls, and the test area checks on every form that nothing was sent |
| What a page throws away is noticed | After writing, EMPTY says which fields hold nothing; those are written again the way typing arrives; what is still empty is reported to the person, not counted as filled |
| A choice is made only when something settles it | form_choices.settle uses the person's details and the job; one call to Claude covers the rest, and an answer that is not one of the field's own options is dropped. Every choice is marked for checking |
| Only the assistant's own answers are read back | To learn the person's corrections, VALUES is asked for the numbers of the fields the assistant filled as answers, and for nothing else |
| Only empty fields are written | The filling script tests each field again just before it writes |

Continuation of Table 6.4

| Rule | How it is kept |
|---|---|
| Values are data, never code | They travel as JSON inside the script, and the script travels as an argument of osascript |
| What the page answers is not trusted | read_answer and fill_answer accept only the expected shape and cut every text to a fixed length |
| Private fields are left alone | Passwords, payment and personal-status fields are recognised by their kind or their label and never planned |
| A file is attached only when it is plainly the right one | The field must ask for a CV, a letter or other attachments, hold no file, and take the kind of file; the job must be known. A photo, a passport or a transcript is never attached. The page must report a file of the same size, or the field is listed as left |
| The right page gets the values | The filling script compares the page's address with the one that was read, and writes nothing when they differ |
| The browser is never started | The request first asks whether the browser is running and stops there when it is not |

## How the assistant decides what to do

Figure 6.6 shows what happens to a request.

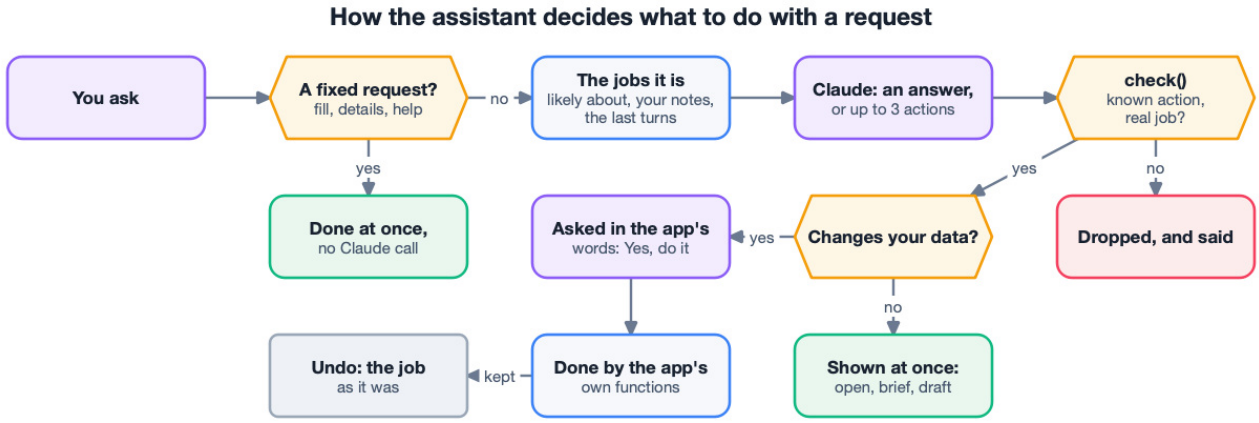



Figure 6.6 – How the assistant decides what to do with a request

A request that is not one of the fixed ones ("fill in the form", "my details", "help") goes to assistant_brain.decide: one call to Claude, with the jobs as lines, the

numbers, the last turns of the talk and the list of actions. Claude answers in words or names one action. Table 6.5 gives the rules that keep this safe; api_talk carries them out.

Table 6.5 – The rules of acting on a request

| Rule | How it is kept |
|---|---|
| Claude does nothing itself | It returns JSON. check keeps only words, up to three actions from a fixed list, and arguments of the right kind; an unknown action or a job that is not in the list is dropped |
| Only the jobs a request is likely about are sent | relevant picks the job in front, the companies named in the talk, what is open and the best leads, up to 70; closed applications come only when the request is about them. Sending every job with every request cost three times as much |
| What was agreed to can be taken back | assistant_confirm keeps the job as it was, and any lead that marking it applied folded away; assistant_undo puts them back. Not offered while a search is running |
| The prompts were tried against the real Claude | A trial on invented data showed a reply's date being read as an interview's date, and a guessed reason for a score. The job lines were reworded ("an invitation arrived", "no date set") and the rules tightened; run such a trial again after changing a prompt |
| Nothing of the person's is changed without their yes | An action that changes data comes back to the window as a question, worded by the app from the checked arguments, not by Claude. assistant_confirm carries it out and checks it once more |
| A rewrite is shown before it is saved | The new paragraph or answer is made first and travels with the question; saving is the confirmed action |
| An answer comes from what the app holds | The prompt carries the jobs, their state and the facts about the app; a brief, advice, a review and practice feedback each have their own prompt with only their own facts |
| The fixed requests cost nothing | They are recognised by their words before any call is made |

Continuation of Table 6.5

| Rule | How it is kept |
|---|---|
| Initiative uses no model | The briefing and the reminders are computed from the data; noticing a form is off until it is turned on, and looks only at the address and title of the tab in front and the number of its empty fields |

### How the window talks to the logic

Figure 6.7 shows the way of one press of a button through the layers and back.

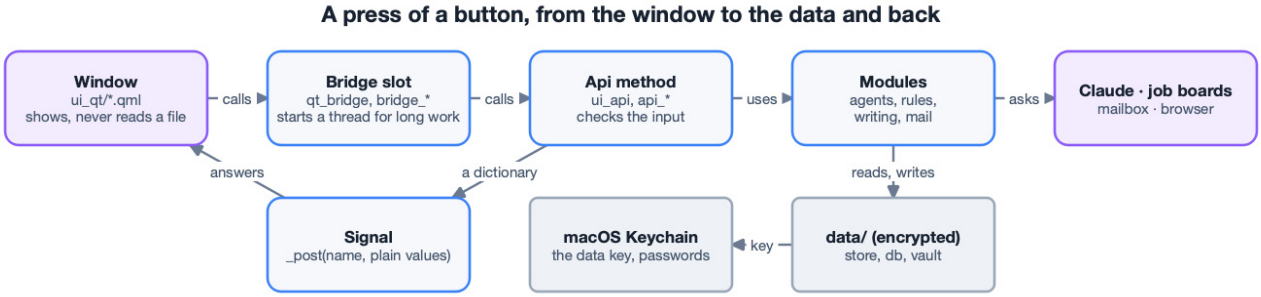



Figure 6.7 – A press of a button, from the window to the data and back

The window never touches a file. Every action goes through three layers:

1. A QML file in ui_qt/ calls a slot of bridge, for example bridge.saveOptions(...).
2. The slot, in qt_bridge.py, bridge_runs.py or bridge_jobs.py, calls a method of Api.
3. The method, in ui_api.py or one of the api_*.py files, checks the input, does the work and returns a plain dictionary: {"ok": True, ...} or {"ok": False, "msg": "why"}.

Long work (a search, a scan, a rewrite) runs in a thread and answers through a signal.

> **Important**
> Every input is checked twice: in the window, where the person sees a red message next to the field, and again in the Api method, so that nothing unusable is saved whatever sent it.

## 7. Work with the data files

Figure 7.1 shows how the data is kept safe: every read and write goes through vault, the key lives in the Keychain, a backup is made before a bulk change, and a rule file changed outside the app must be approved.

Everything a person makes with the app is in data/ (Table 7.1). The folder is not part of the code and is never committed.

Follow these rules when code changes data:

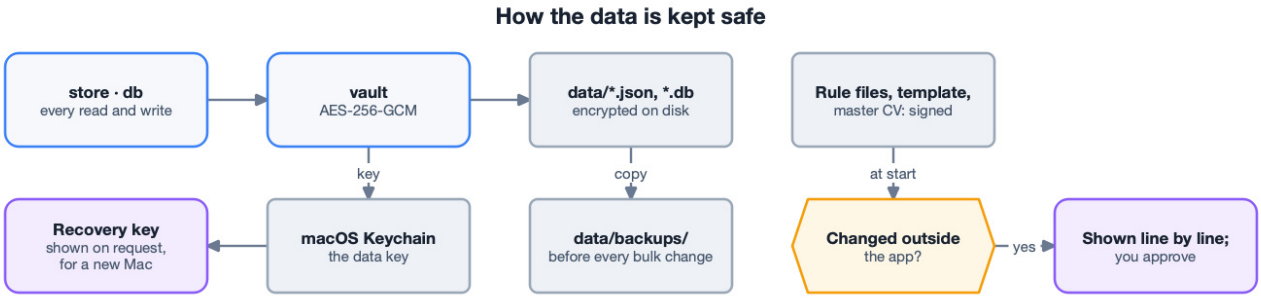



Figure 7.1 – How the data is kept safe

Table 7.1 – The files and folders in data

| Path | What it holds | Encrypted |
|---|---|---|
| data/jobs.json | Every posting and application | Yes |
| data/profile.json | The profile read from the CV | Yes |
| data/ui_settings.json | The Options | Yes |
| data/responses.db | Every reply that was read | Yes |
| data/usage.json | The Claude usage and the last known limits | No |
| data/backups/ | Backups, made before any bulk change | Yes |
| data/cv/, data/my_letters/ | The master CV and the person's own letters | No, so that Word can open them |
| data/tailored_cvs/, data/cover_letters/ | The documents the app wrote | No, so that Word can open them |
| data/app_trouble.log | One line for each error the app carried on after | No |

1. Read and write the job list only through store, the replies only through db, and any encrypted file only through vault.
2. Call vault.make_backup(label) before a change that touches many jobs.
3. Move a document to the encrypted backup instead of deleting it; put old copies in the Trash.
4. Do not change the data while the window is running.

# 8. Test a change

## Run the tests

1. Run the whole suite. It takes about two minutes and needs no network:

   ```
   python -m pytest -q
   ```

2. Check for unused and undefined names:

   ```
   python -m pyflakes jobhelper tests scripts
   ```

Both must pass before a change is committed.

## What the tests may never touch

The tests work in temporary folders with stand-ins. Table 8.1 lists what they never reach and what stands in for it.

Table 8.1 – What the tests never touch

| Never touched | What stands in |
|---|---|
| The real data/ folder and .env | Temporary folders, set by the isolate_data fixture |
| The macOS Keychain | An in-memory keychain |
| Claude | A function that returns a prepared answer |
| Job boards and mail servers | Functions that return prepared postings and messages |
| Notifications, Finder, the Trash, the browser | Functions that only record that they were called |
| The person's browser | A guard that fails the test. The scripts that read and fill in a form run in a browser engine on a local page, tests/fixtures/forms/ |

## Test the window

The window is loaded without a screen and driven by the tests in tests/test_ui_qml.py. Use the helpers of its ui fixture: ui.js(code) runs JavaScript in the window, ui.find(name) finds an item by its objectName, and ui.wait(ms) lets the window run.

> **Tip**
> After a change that can be seen, take a picture of the window and look at it. A test that reads a text does not notice that the text is not drawn.

# 9. Update the documentation

Each change that a person can see needs three things: a line in CHANGELOG.md, the

user guide brought up to date, and its pictures taken again if a pictured window changed.

### Take the pictures of the user guide again

The pictures show the real window filled with invented people and companies. A script takes them:

```
python -m pytest tests/guide_shots.py -q -o addopts=""
```

### Keep the numbers of figures and tables in order

Figures and tables are numbered inside their chapter, as in a thesis: the second figure of chapter 4 carries the number 4.2. Every picture has a caption under it and every table a caption above it: a line in italics made of the word Figure or Table, the number, an en dash and the title. The text refers to each by its number.

1. Write the caption of a new figure or table with any number that is not used yet, for example 900.
2. Renumber the captions and every reference in one go:

   ```
   python scripts/guide_number.py
   ```

### Build the PDFs

Figure 9.1 shows how a PDF is made from its text.

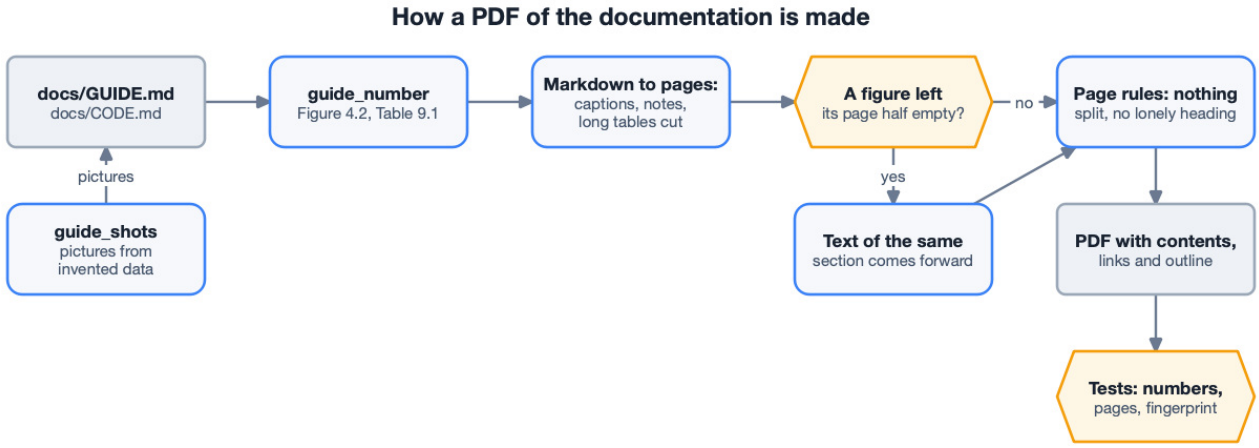



Figure 9.1 – How a PDF of the documentation is made

```
python scripts/guide_pdf.py
```

The script makes docs/GUIDE.pdf and docs/CODE.pdf. It lays each one out like a thesis: a title page, the contents with dotted lines that lead to the page numbers, then the text with its page number at the top right. The entries of the contents, references to figures and tables, and links to another chapter can be selected and lead to their place. A caption stands centred under its picture and above its table. There are no lists of figures and tables. The script keeps the page rules in Table 9.1. A test fails while a PDF is older than its text.

Table 9.1 – The page rules of the PDFs

| Rule | How the script keeps it |
|---|---|
| A figure is never split across two pages | A picture and its caption are one block; a block that would cross a page starts on the next one |
| A table stays on one page | The same, with the caption above it |
| No page is left half empty because a figure or a table had to start on the next one | The text that follows it in the same section is brought forward to fill the space; a figure, a table or a note comes forward only if all of it fits. Text never crosses a heading. A table longer than a page begins where its page has room, and a picture that almost fits is shown a little smaller |
| What names a place leads to it | A PDF library adds a link for every entry of the contents, every mention of a figure or a table, and every link to another chapter, and writes the chapters into the PDF's outline |
| The points of a list stand out | A heavy square before each point, a bold number before each step |
| A table longer than a page continues between two rows | It is cut into parts that fit, each under the line "Continuation of Table" with its number, and with the heading row repeated |
| A heading is never left alone at the bottom of a page | A heading whose text would start on the next page moves there too |

### Write in the house style

Table 9.2 gives the rules for the text of both documents.

Table 9.2 – The rules for writing the documentation

| Element | Rule |
|---|---|
| Headings | Start with a verb and name the task: "Connect your mailbox" |
| Steps | Numbered, one action each, in the order they are done |
| Reference material | In a table, with a caption and a reference from the text |
| Names of buttons and labels | Bold, spelled exactly as in the app |
| Commands and file names | Monospace. The user guide contains no commands |
| Links | The link text says where it leads, never just "here" |
| Notes | Four kinds: Note, Tip, Important, Warning, as a quoted block that starts with the bold word |

## 10. Troubleshoot problems

Table 10.1 lists the problems met most often while working on the code.

Table 10.1 – Problems while working on the code

| What you see | Cause | What to do |
|---|---|---|
| "Unset ANTHROPIC_API_KEY to use your subscription" | The paid key is set in the shell | Run unset ANTHROPIC_API_KEY, then start again |
| The window says Claude is not signed in | The Claude Code program has no session | Run claude auth login |
| A test of a PDF fails after a change to a document | The PDF is older than its text | Run python scripts/ guide_pdf.py |
| A test of the numbering fails | A figure or table was added or moved | Run python scripts/ guide_number.py |
| A QML property named onSomething does nothing | QML reads such a name as a signal handler | Give the property another name |
| An item made by a Repeater is not found in a test | It is not a child in the object tree | Use ui.seen(part) or ui.drawn(name) |
| The window stutters while a thread works | Python runs one thread at a time | Give the thread short rests, as keywords.read_ahead does |
| Data looks wrong after an experiment | A change was made to the real data | Restore the newest backup: python status.py security restore |